Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. We trained each model for 30 epochs with a batch size of 128.

2 Results

Figure 1 shows the test accuracy of EfficientNet by dataset.

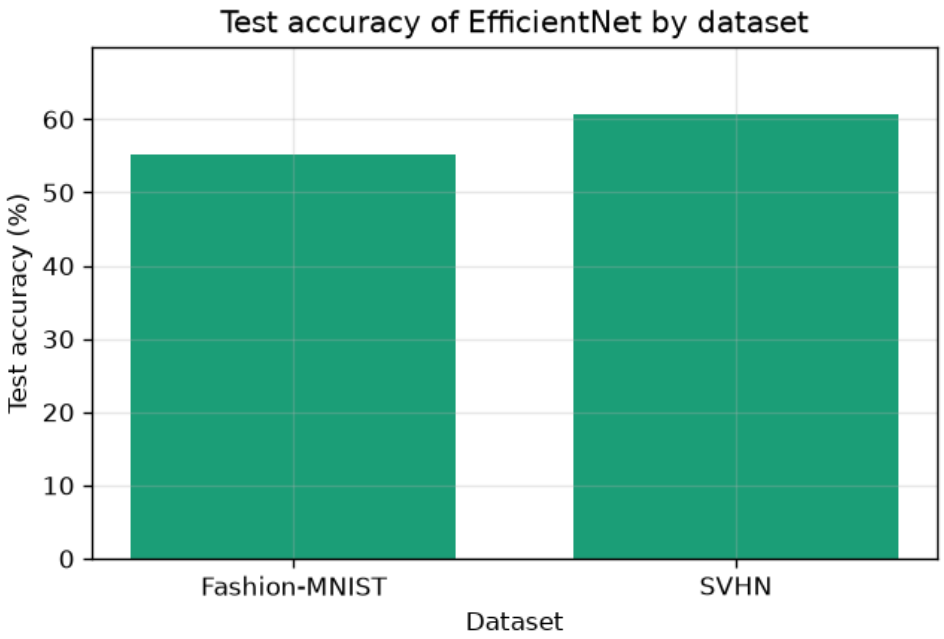


Figure 1: Test accuracy of EfficientNet by dataset for the models in this experiment.

On SVHN, EfficientNet obtains a test accuracy of 95.6%. EfficientNet reaches 55.2% test accuracy on Fashion-MNIST. EfficientNet reaches 61% test accuracy on SVHN. EfficientNet achieves 55.2% test accuracy on Fashion-MNIST.

Figure 2 shows the accuracy during training.

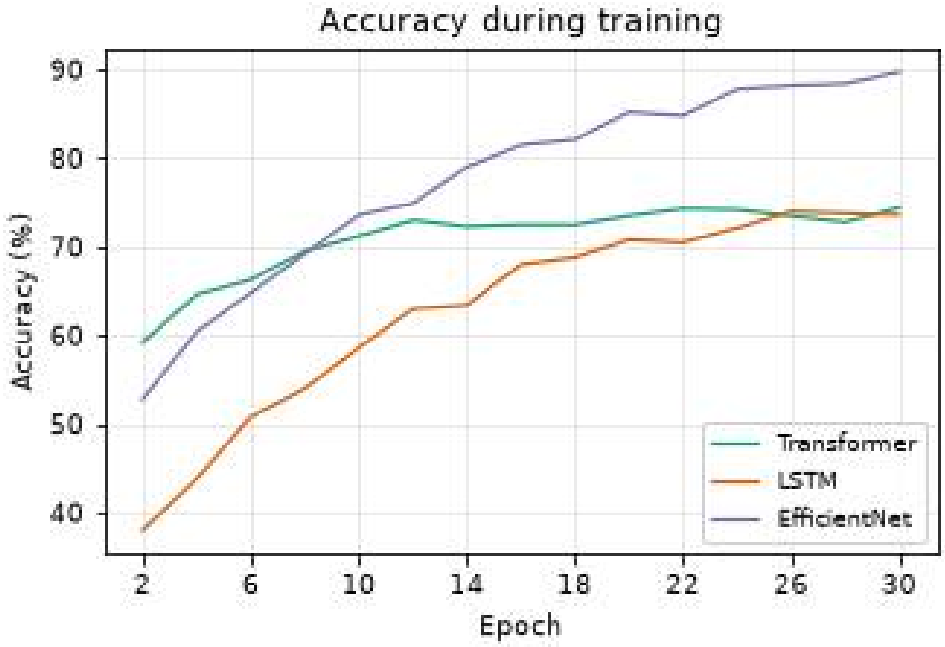


Figure 2: Accuracy during training for the models in this experiment.

EfficientNet attains a accuracy of over 85% at epoch 30. At epoch 16, EfficientNet is 11.8 points better than Transformer.

Figure 3 shows the f1 score during training.

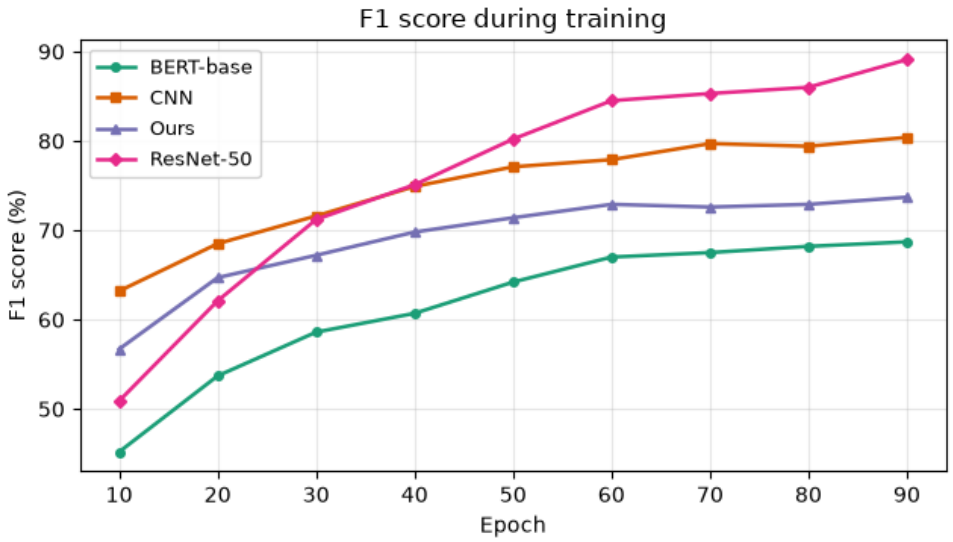


Figure 3: F1 score during training for the models in this experiment.

At epoch 40, Ours achieves 68.8% F1 score. At epoch 80, Ours is 28.8 points better than BERT-base. At epoch 10, CNN achieves over 60% F1 score (Figure 3).

3 Discussion

On STL-10, SVM obtains an accuracy of 74.1%. Overall, the results are in line with what we expected from the literature.